



Milestone 5

Critical Design Evaluation

Bring ROAR to Life - Team 1

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EML 4502 – Senior Design 2

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I. Introduction

The end goal of Bring ROAR to Life is to design and create an anatomically accurate Stegosaurus that is able to walk and interact with its environment in a lifelike manner. One of the primary challenges in this project is due to how open ended the initial requirements are, which has given the team a lot of creative freedom on where to place priorities. These have driven our primary requirements to focus on visual aesthetics and biologically representative walking.

On our journey in this project, we have been in contact with both the Lawrence Hall of Science at Berkley, along with Dr. Matthew Dempsey, a musculoskeletal biologist specializing in dinosaurs and more recently the Stegosaurus. We have been in contact with both parties extensively in our research for this project in order to better define our requirements and their input to date has been invaluable.

The following sections will present the current state of the project and the research completed to date. From this report, we will be able to begin the prototyping process of our dinosaur. Our end goal of an ideal project will be to provide an educational resource for museums in general. However the main goal of this specific project will be to provide a base functional prototype. And to possibly add value to the Lawrence Hall of Science through showcasing the design process in an interactive way. We will discuss the ways an end product may fail, mechanical modeling, gait and programming analysis, along with a discussion on packaging and manufacturing.

II. Failure Modes and Effects Analysis

In terms of our project, there are a lot of moving parts and variables to consider. A full table is given in Appendix A. There are seven major areas to consider for this project: balance, structure, power, joint articulation, control systems, safety, and environment.

The robot must be able to balance both statically and dynamically. Any case of the robot falling over or not being able to handle its environment needs to be considered as it may lead to a system or sub-system failure. Its main structure also must be able to handle the weight and reaction forces of the robot as it moves.

The whole robot must be powered by the same power system, and all constituent components cannot cause the power system to become overburdened. There is a real risk when using a rechargeable lithium battery that charging speeds, discharging speeds, and possible punctures can have an adverse reaction on the health of the battery and possibly start a fire. Special care must be taken in this regard when it comes to consumer safety.

Our robot is also heavily focused on having realistic movement, which depends upon the accuracy and reliability of the joints and articulation systems. If those systems are inaccurate or in their operation damage the robot or something in their environment, this poses a problem. The control systems of the robot are the center of anything and everything actually working. Operators of the robot must have full control over the robot at all times and the robot's detection systems must be calibrated properly and send proper signals to the robot's microcontroller. Errors or defects in this system could cause problems with balance, action, navigation, and realism. The robot *must* shut off if communication with the controller is lost at any time. And will be discussed further later in this report.

Lastly and most importantly is the safety and operational environment of the robot. At all times, the safety and well-being of both the operators of the robot and the general public is of paramount importance. For proper operation of the robot, the intended operating environment must be respected, as extreme heat, cold, and wet weather or surrounding conditions (i.e. a pool, puddle, lake, or any body of water) must be strictly avoided for proper operation of the robot. The robot is not designed to be struck or fall in any significant way, doing so may damage the robot permanently.

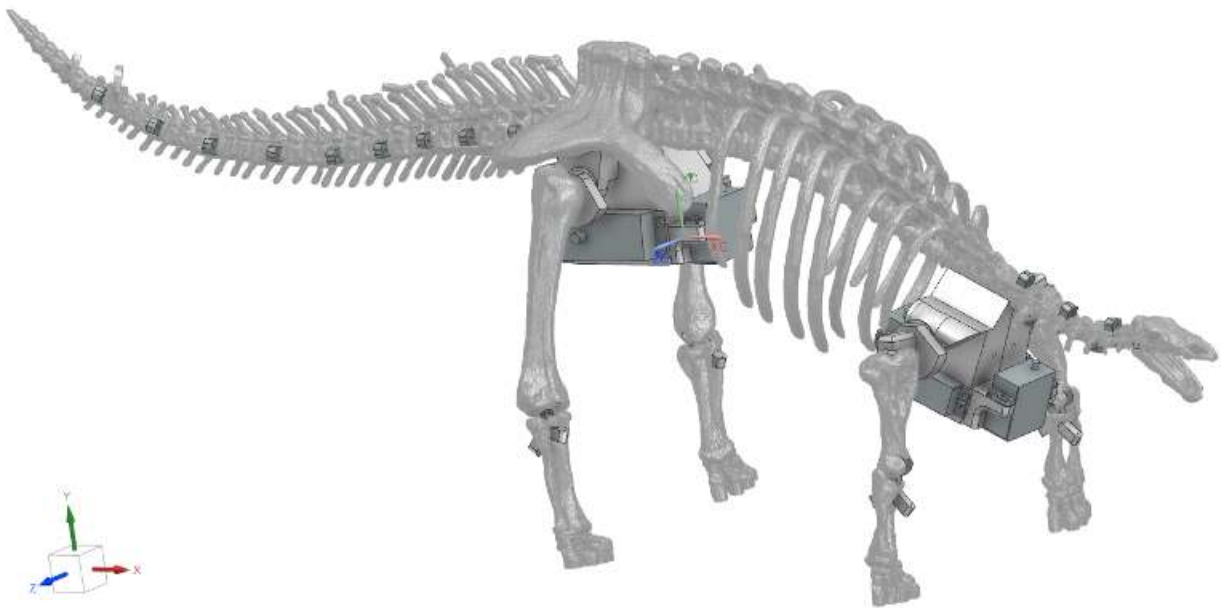
In general, the robot is intended to be operated in an air-conditioned, indoor space, by a trained operator. It is not meant to carry loads of any kind, nor to be used as a toy and handle things like falls or hits. It is not waterproof, and should have monitored interactions with the general public.

III. Modeling

To get an idea about how a final product may take shape, it is critical to make a detailed model for many aspects of the robot. This not only includes a 3D CAD model, but also includes structural analysis of a few components, a detailed gait study to understand how to program and make the robot move smoothly, and electrical schematics to understand how to wire and make everything move.

3D Modeling

To effectively 3D model our prototype this will primarily take place in Siemens NX, this was primarily chosen based on the groups extensive experience with NX, but it is also preferred as it is able to handle large assemblies effectively. Below is the full model as completed for M5 there are a few minor updates that will be required for M6 but all primary joint types and structure has been completed.



**Figure 01: Current Layout*

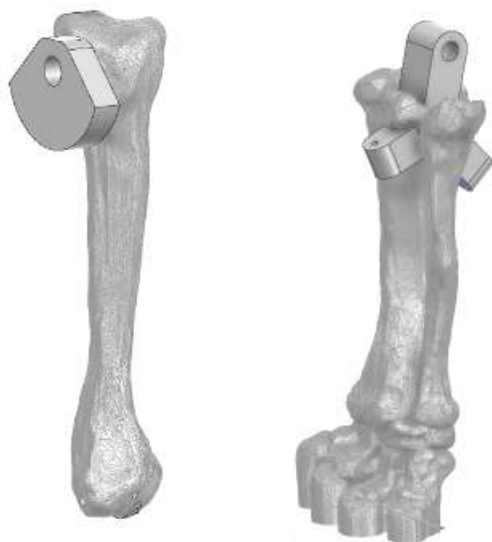
Joint Types:

The final joints for our prototype have remained generally consistent with the conceptual design analysis in M3. There are still a few minor things that will need to be addressed such as the rear femur drive wheels, as there is not an accurate model of the servo accessories and arms that this drive wheel will need to be attached.

Leg joints will use capstan drives for the femur and upper arms, this was chosen for low slop if tuned correctly, while being able to implement an effective gear ratio. Knees and Elbows will use internal cable routing within the existing bone structure. This will effectively provide a direct drive from the servos but allow for them to be remotely mounted.

Rear Legs:

The rear legs will consist of a femur and tibia. The integrated capstan drive on the femur will allow for an integrated gear reduction. These will have internal cable routings that will pick up from the tibias in order to mount the motors remotely in the larger hip structure.



**Figure 02: Rear Femur (Left) and Tibia (Right)*

Hips:

The rear hips are based off of a single piece and are mirrored to get 2 symmetrical sides. This is a similar philosophy to what was used for left and right legs, and allows for simplified manufacturing, and to maintain a symmetrical dinosaur as the skeletons available have been slightly warped over millions of years. Both the femur and tibia drive motors are housed in a lower mount.



Figure 03: Rear Hips

Front Legs:

The front legs will follow a very similar design language as the front legs. These will use capstan drives for the upper arms, and an integrated cable for the forearms. The mounting type and bearing choices will be carried over from the rear in order to simplify the design and reduce service part count for both manufacturing and future use.

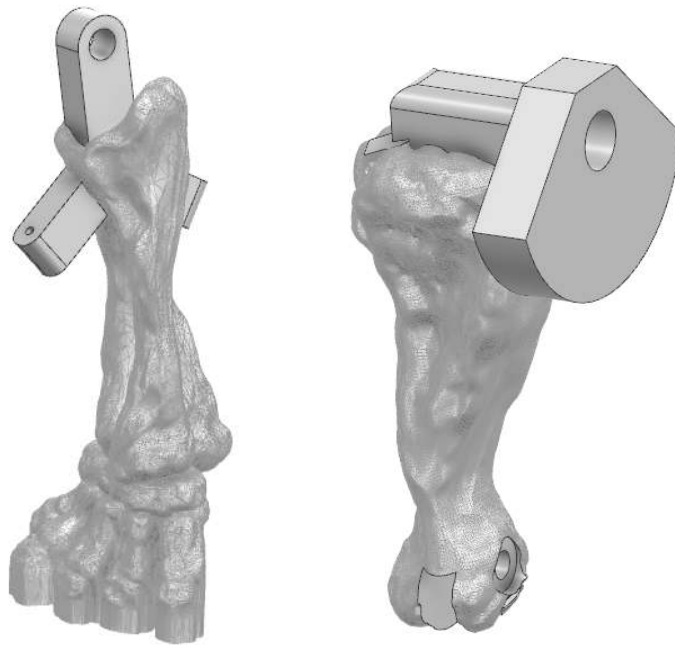


Figure 04: Front Forearm (Left) and Upper Arm (Right)

Front “Shoulder Blades” & Mounting:

To mount the front legs, this will utilize a similar design to the rear hips, the main difference being the lack of a bony hip structure, and the drive motors for the upper arms will utilize the medium sized 30kg/cm servos. As the overall leg length is significantly shorter than the front, and will require a lower torque to effectively walk.

For mounting, the current plan is to use an epoxy to attach the front leg mounts to the existing spine, and include a bracket that locates the front legs in the correct location. For the forearms, there will be integrated cable routings in the arm mount similar to the rear hips, with the motors attached off the sides.

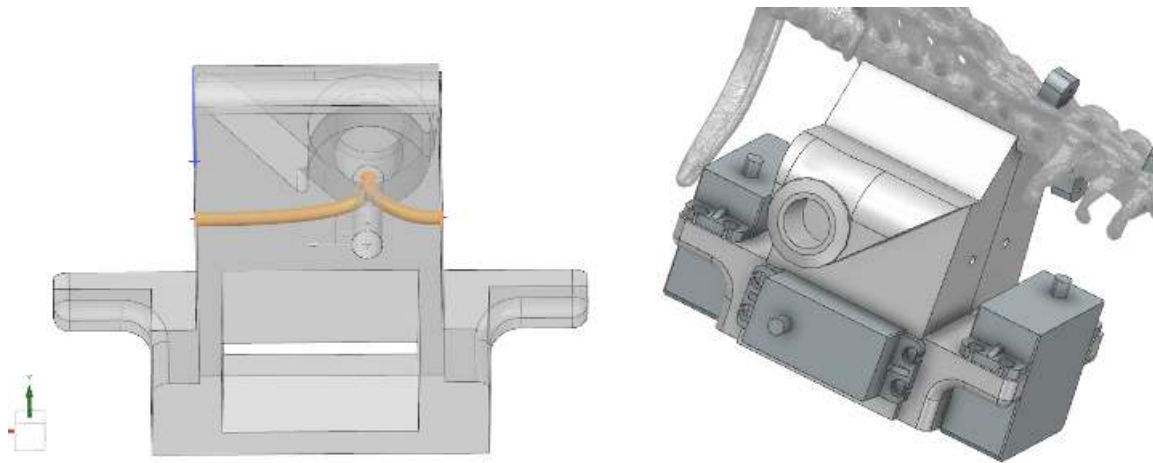


Figure 05: Front Motor Mount

Rib Cage:

The rib cage will house the majority of the electronics, this will allow them to be hidden from the main view, and covers will be made to make the electronics and wiring less obvious. When compared to an existing stegosaurus skeleton, this design has been simplified with a lower overall rib count, contributing factors can be seen in the rear hip motor mounts and there were 3 ribs on each side that needed to be removed to mount the front motors. Additionally there are some duplicated ribs, this will allow for them to be simpler to manufacture and replace, while having a minimal impact on the visual accuracy of the dinosaur.

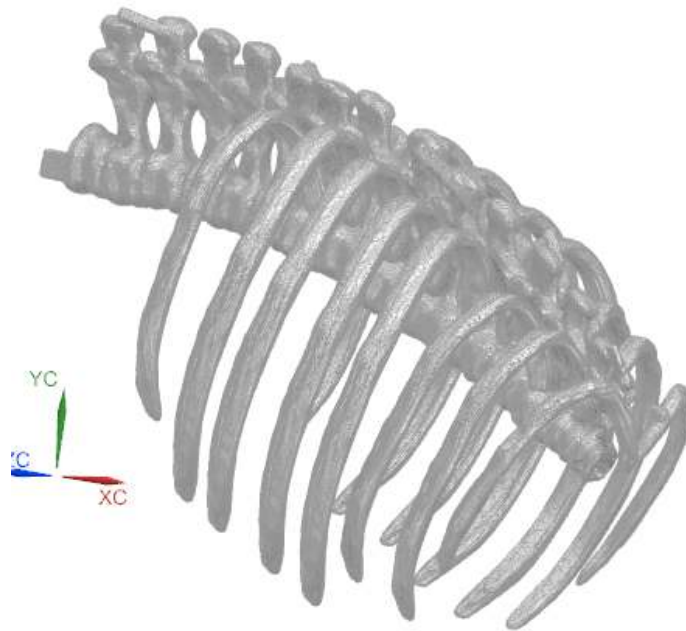


Figure 06: Ribcage

Hardware and Manufacturing

Most of the parts in this robot will be 3d printed. If a prospective product were to come to market, then injection molding, machining, and possibly resin printing may be used in order to increase the surface finish and overall quality of parts.

For some of the leg members, 3d printing will be the only way we will be able to create features such as internal routing for cables in a relatively small part. The leg joints will also need to be supported. In this case we could use anything from plastic on plastic joints, to bushings, bearings, and more. The top joint on each leg will be supported by needle bearings to create smooth motion, the knees and elbows will first use a plastic on plastic joint with a screw acting as a pin. This may need to move to a steel or metal rod end to better support the lower leg members.

Electronics Packaging

The housing for the electronics is approximately 5.2in. x 4.5in. x 3in. Strategic placement and orientation of the components in the housing is essential. Many components are fitted precisely into a tight space. This housing will fit under the ribcage; the cover will be connected to the dinosaur's spine, and the body of the housing will be connected to the cover via heated inserts and bolts. Some electronic components have slots and will be bolted into the housing via heated inserts, and some of the electronic components do not have slots, and will be free floating in their designated space. To avoid inconsistencies and damage, all cables must have a minimum bend radius of five times the overall cable diameter, which is a safe standard rule of thumb when working with wires. Appropriate space was left for the wires, and cutouts were made so that

wires can go out of the housing and into the motors. The housing will be 3D printed upright and will use supports where needed. Tolerancing and spacing is accounted for when it comes to cutouts and holes for heated inserts. Fillets were placed to strengthen the housing structure.

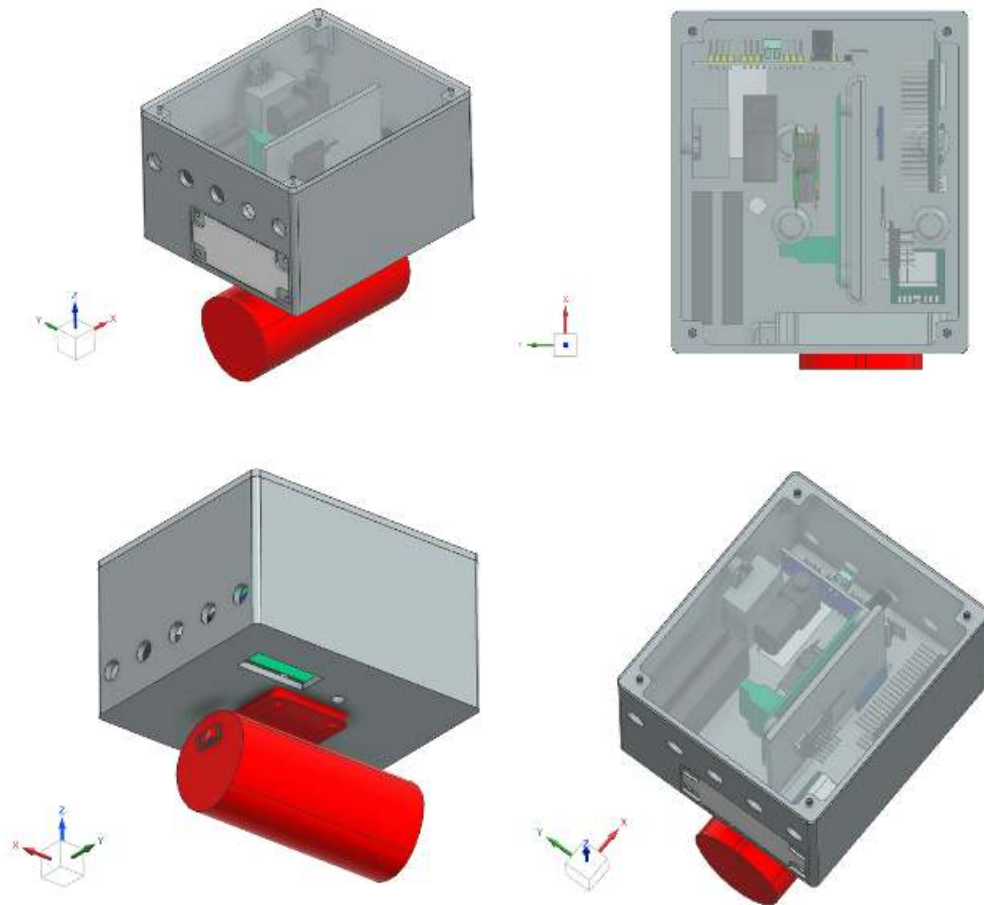


Figure 07: Electronics Housing

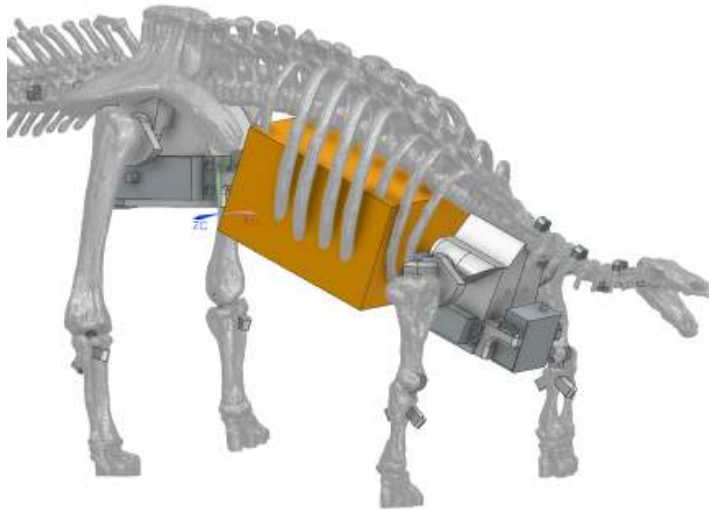


Figure 08: Electronics Housing (orange) Realestate

IV. Analysis

Gait Study

A gait study provides data on how a creature moves. For the purposes of this project, a gait study will provide the angular position of each joint. This information will then be taken by the programmer and incorporated into the control system. The control system will send signals to the motors to move each joint to a specific angle at each point during a walk cycle.

Due to the extinct nature of the stegosaurus, there is no precise information on the joint angles of the animals walking pattern. The team analysed the skeletal structure of the dinosaur to determine the maximum range of motion of each joint. This gave the outer limits of the range of motion. The team then utilized tabulated information on the gaits of other animals, particularly humans, to create angle lists for each joint with the correct cadence. An animation using MATLAB was created to visualize the walk of the animatronic. From there, edits could be made to the angle lists. The angle data was then handed over to the programmer to be integrated into

the control system.

Observing the stegosaurus skeleton, the rear leg appears to have a maximum range of motion of 85 degrees. The front leg does not move further than 15 degrees past the vertical, while having a much more range of motion backwards. An animation resource was used to pinpoint key joint angles throughout the strides []. The key joint angles are captured in the tables below. The joint angles were averaged between each point in the stride to create a smooth transition.

Table 1: Front Leg Joint Angles			
Stride Status	Gait Percent	Hip Angle	Knee Angle
Heel Touch	0%	135°	150°
Full Load	50%	115°	210°
Toe Off	65%	105°	150°
Leg Swing 1	75%	70°	100°
Leg Swing 2	85%	120°	95°
Heel Touch	100%	135°	150°

Table 2: Rear Leg Joint Angles			
Stride Status	Gait Percent	Hip Angle	Knee Angle
Heel Touch	0%	75°	160°
Full Load	50%	90°	175°
Toe Off	65%	100°	160°
Leg Swing 1	75%	110°	150°
Leg Swing 2	85%	95°	130°
Heel Touch	100%	75°	160°

Once the angles were averaged, the angles were passed through a function to smooth the sharp transitions further. The resulting joint angles are portrayed below. The joints of all the limbs point in towards the center of the body. This is noted in the shoulder and hip joints. The

elbow and knee joints have similar paths, though the knees have a peak just after the 50% mark of the gait that is lacking in the elbow stride. The elbow has a sharper trough than the knee. The hip and the knee have mirrored paths that are near identical.

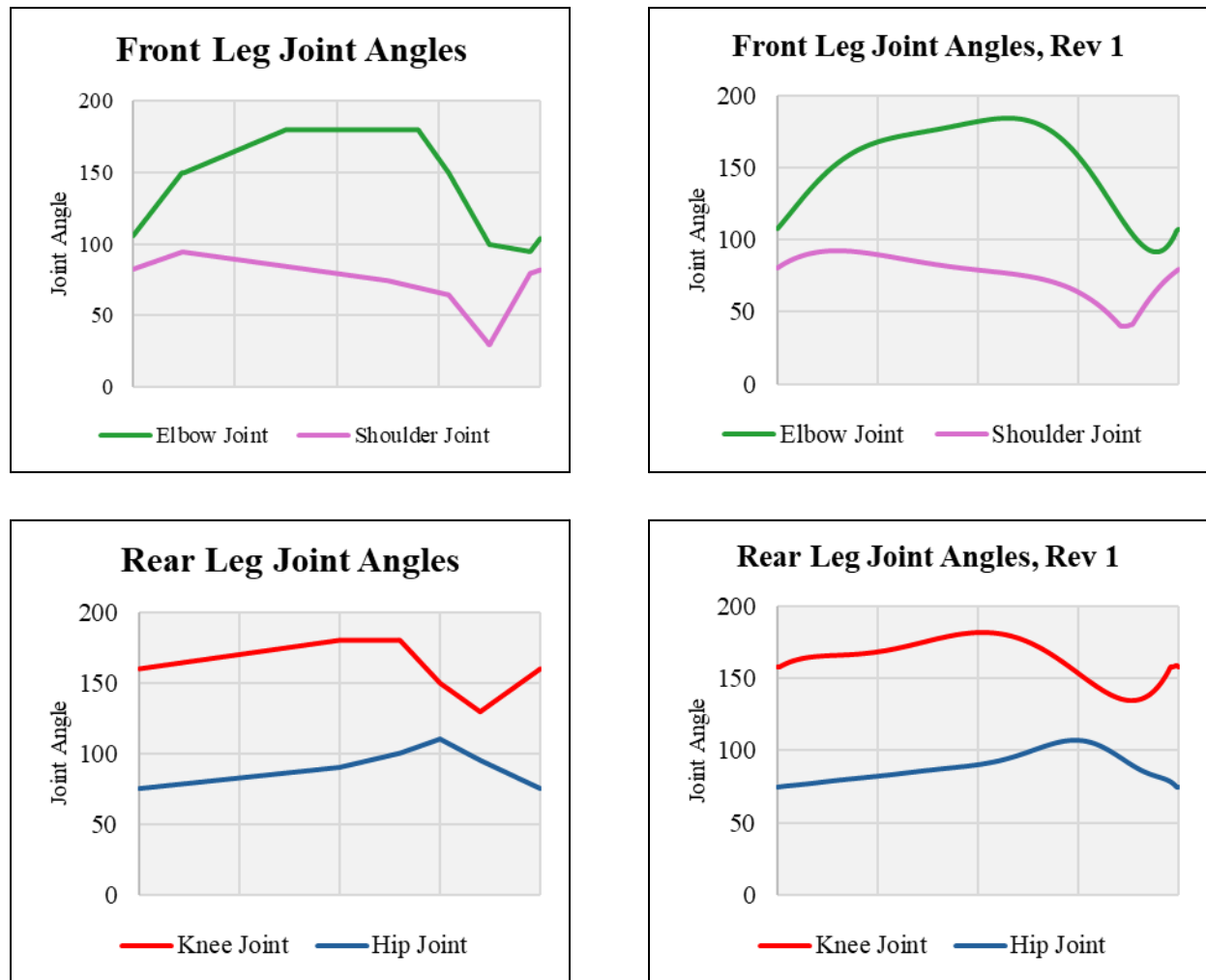


Figure 10: Joint Angle Data

The tabulated joint angles were added to MATLAB to be animated. The code included two modifying variables, one to change the overall speed, and one to turn the animatronic. The speed is adjusted by the frame rate. Increasing the frames per second easily allows the animatronic to walk faster. To turn the animatronic, a multiplier was applied to the left and right side of the

body. The multiplier adjusts the joint angles evenly between the two sides of the body. If one side of the body has a narrower range of motion than the other, it will allow the body to turn.

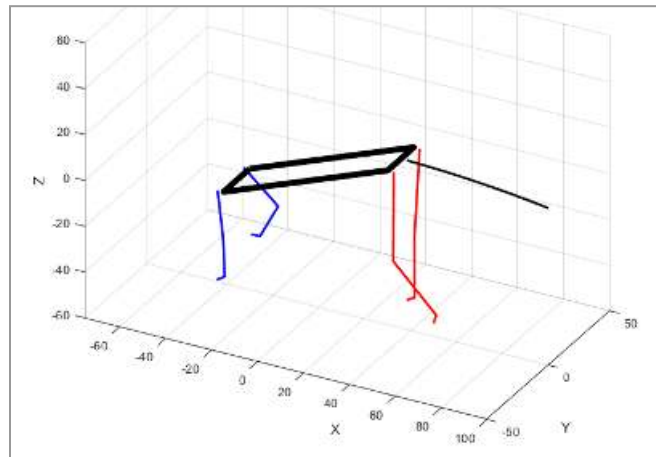


Figure 11: MATLAB Animation of the Gait

An interesting point of comparison exists between a human gait and the stegosaurus's gait. The human hip has a range of 50 degrees, where the stegosaurus has a range of motion of the front shoulder of 60 degrees and a hip range of motion of 35 degrees. However, the human hip has a longer continuous change in motion, whereas the range of motion of the stegosaurus hip is brief but sharp. The human knee features two sharp peaks in joint angle. The stegosaurus knees and elbows have one strong peak with one minimal peak. In general, the path of the human knee joint is very similar to that of the stegosaurus, while the hip differs dramatically. This follows with the anatomy of the human and stegosaurus. The bipedalism allows for the human hip to be more flexible, whereas the stegosaurus has more stable hip and shoulder joints to support its four limbs. The knees and elbows, by comparison, are anatomically similar.

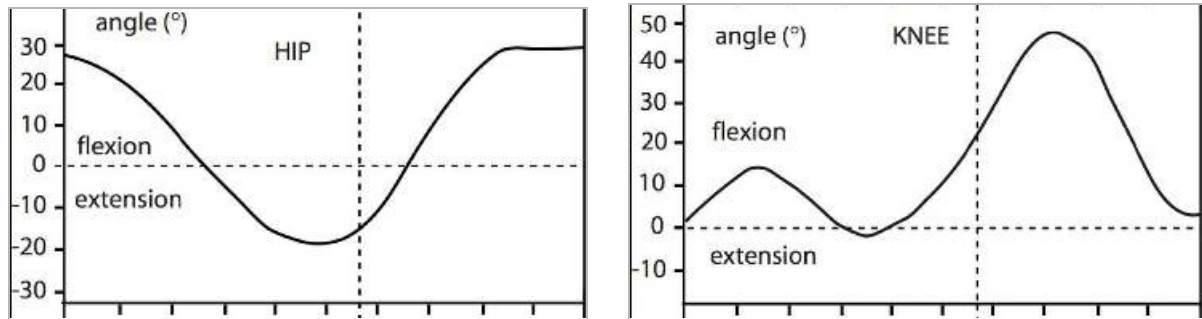


Figure 12: Human Joint Angles [1]

Torque Calculations

Below are the results of the torque calculations. When the animation was run, the angular position was integrated to provide angular acceleration. This was done at 100 frames per second, where the gait is set into 200 frames. Therefore, each leg of the stegosaurus was taking one step every two seconds.

Table 3: Torque Data				
	Front_Femur	Front_Tibia	Rear_Femur	Rear_Tibia
Max Absolute Angular Acceleration	8298.22	9326.6	6518.8	11216.47
Length (in)	4.0000	3.4000	7.8200	5.2800
Mass (lb)	1.19E-01	1.01E-01	2.33E-01	1.58E-01
Max Possible Torque (lb*ft)	3.06	2.11	17.954	9.51

The maximum absolute torque was found. Then, using the lengths of each limb and the mass-per-volume of the modeled parts, the maximum possible torques were calculated. These torques only provide data on the torques from positional data, but do not include material data. Without the printing being completed for the physical model, the team cannot yet account for torque increases due to friction. One proposed solution is currently to apply a lubricant to reduce friction and improve motor control.

$$I_{yy} = \sum x_i^2 dA_i$$

Formula: Moment of Inertia [2]

$$\tau = I\alpha$$

Formula: Torque [3]

Electrical Schematic

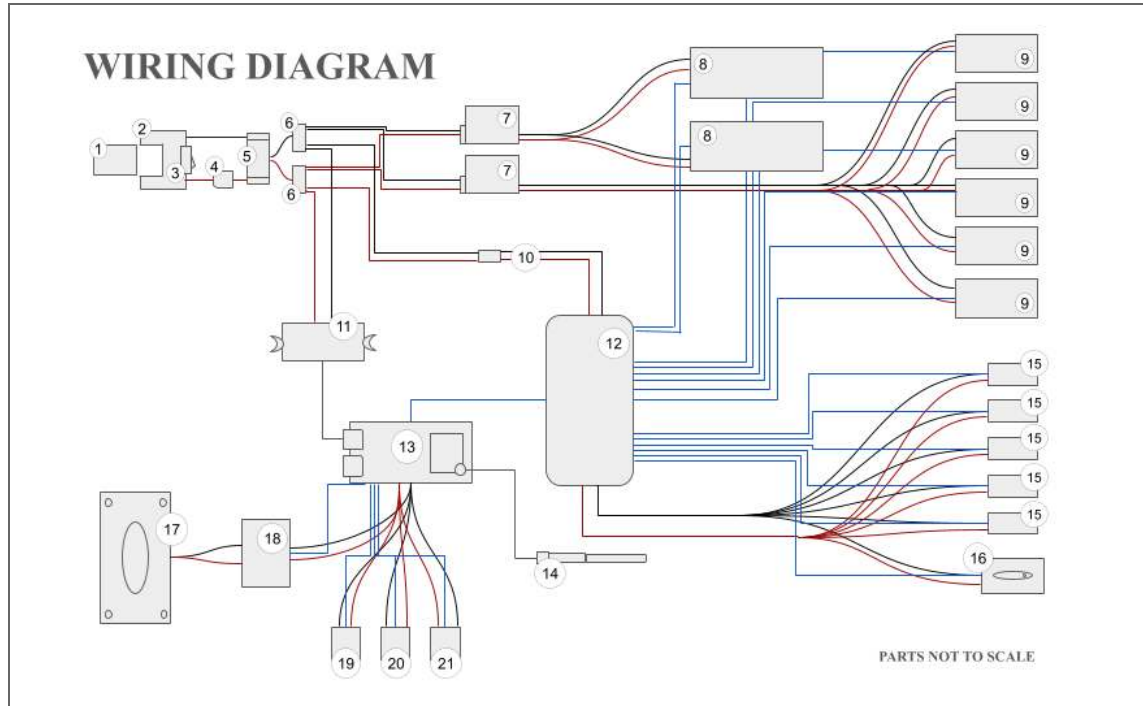


Figure 13: Wiring Diagram

ITEM NO.	SUBSYSTEM	COMPONENT	QTY
1	Power	Milwaukee M12 Battery	1
2	Power	Battery Adapter	1
3	Power	Switch	1
4	Power	Fuse	1
5	Power	Low Voltage Cutoff Board (30A)	1
6	Power	BusBar	2
7	Motor	2-8S LiPo UBEC-8A For 5 Servos Motors	2
8	Motor	DS5180 High-Torque Servo - 80Kg servo	2
9	Motor	DS3230MG Servo - 30Kg servo	6
10	Motor	5V 3A Low Volt Buck Converter	1
11	Motor	5V 3A Buck Converter USB-C	1
12	Motor	PCA9685 16-Ch Servo Driver	1
13	Control	ESP32-S3-WROOM-U1	1
14	Control	Antenna	1
15	Motor	Micro Linear Servo	5
16	Motor	Goteck 1.5g Micro Digital Linear Servo Actuator	1
17	Audio	3W 8Ω Speaker	1
18	Audio	DFPlayer Mini MP3	1
19	Control	MCP9808 Temp Sensor	1
20	Control	GY-NEO6MV2 NEO-6M GPS Module	1
21	Control	Additional GY-521 IMUs	1

This entire system is powered by a Milwaukee M12 battery. This serves as the primary energy source for all subsystems. Power flows from the battery to the battery adapter, then through a main switch, enabling or disabling the entire system manually. After the current passes through a fuse for overcurrent protection, and then goes into a 30A low-voltage cutoff board that prevents excessive battery discharge. From the cutoff board, power is distributed to two busbars that act as central distribution points for positive voltage and ground. All ground connections throughout the system return to the common ground bus, ensuring a shared electrical reference for proper signal operation.

From the busbars, raw battery voltage is routed to two 2–8S LiPo UBEC 8A regulators, which step the voltage down to a stable level suitable for high-current servo operation. These UBECs provide regulated power and ground to the high-torque DS5180 servos and the DS3230MG servos. Additional regulated 5V power is generated by two buck converters: a 5V 3A low-voltage buck converter and a 5V 3A USB-C buck converter. These supply stable logic-level voltage to the control electronics and peripheral modules.

The ESP32-S3-WROOM-U1 serves as the system's central controller. It receives power from a buck converter that is regulated to 5V and shares the common ground reference. The ESP32 communicates with the PCA9685 16-channel servo driver over an I²C interface. The PCA9685, powered from the logic rail, generates PWM control signals for all servo motors. While the servo driver supplies signal lines, the servos themselves receive their operating power directly from the UBEC-regulated supply. This separation guarantees that high-current motor loads do not destabilize the logic circuitry.

In addition to motor control, the ESP32 interfaces with multiple sensors and communication modules. The MCP9808 temperature sensor and the GY-521 IMU communicate with the ESP32 via I²C, sharing the same bus as the servo driver. The GY-NEO6MV2 GPS module communicates over a UART interface. All sensors are powered by the regulated logic supply and share the microcontroller's ground.

The audio subsystem consists of a DFPlayer Mini MP3 module powered from the 5V logic rail. The ESP32 communicates with the DFPlayer via UART to control any audio queues. The DFPlayer directly powers a 3W 8 Ω speaker, providing amplified sound output. The ESP32 is connected to an external antenna, enabling wireless communication with the user's phone and PS4 Controller.

Structural Analysis

FEA is an important tool that will allow us to use predicted loads and see if our base designs will hold up and are in line with the materials we want to use. Below is an example pulled off of the hip joint. This is going to see the highest amount of load by far due to the large motor size, and relatively small support area. Another consideration is in the 3d printing material and layer orientation, which will be further experimented with in this project. For a general idea, the maximum yield strength was set to ABS plastic. At roughly 4000psi, below the finished simulation predicts a max stress of 2320psi in figure 14, which is well under the lower estimate for ABS.

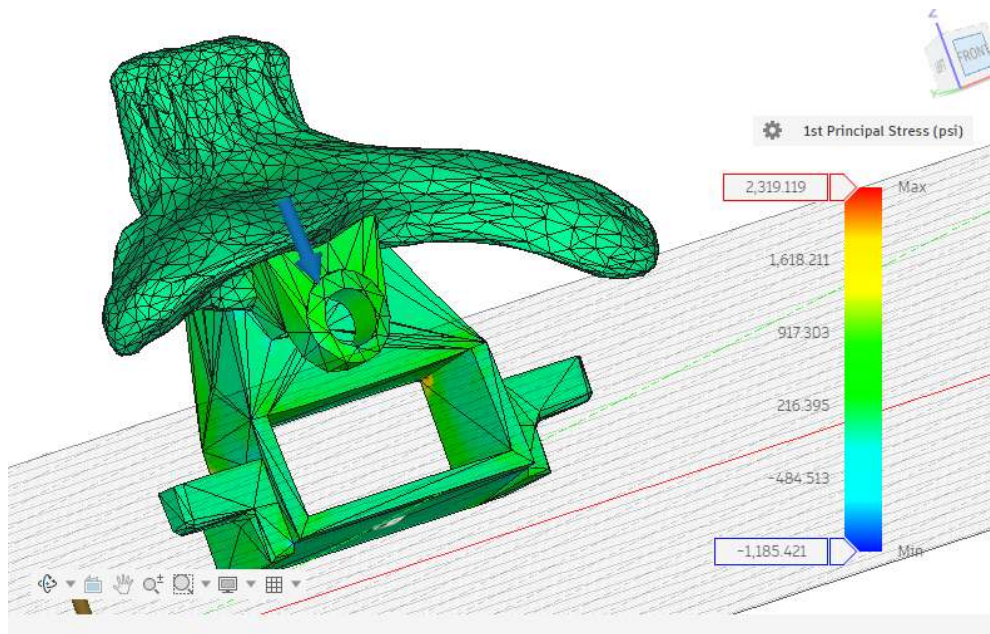


Figure 14: Hip 1st Principal Stress

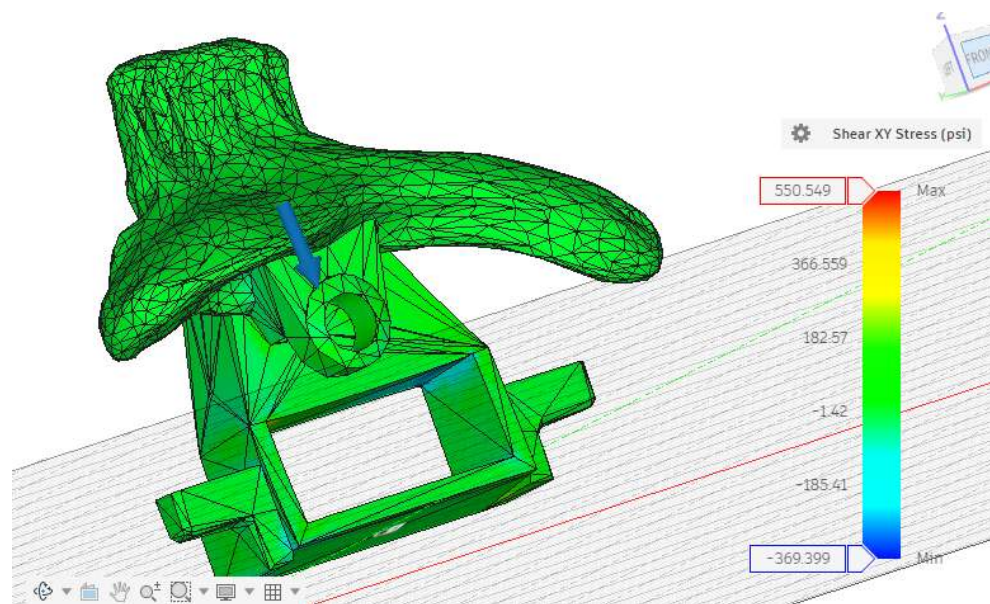


Figure 15: Hip Shear Stress

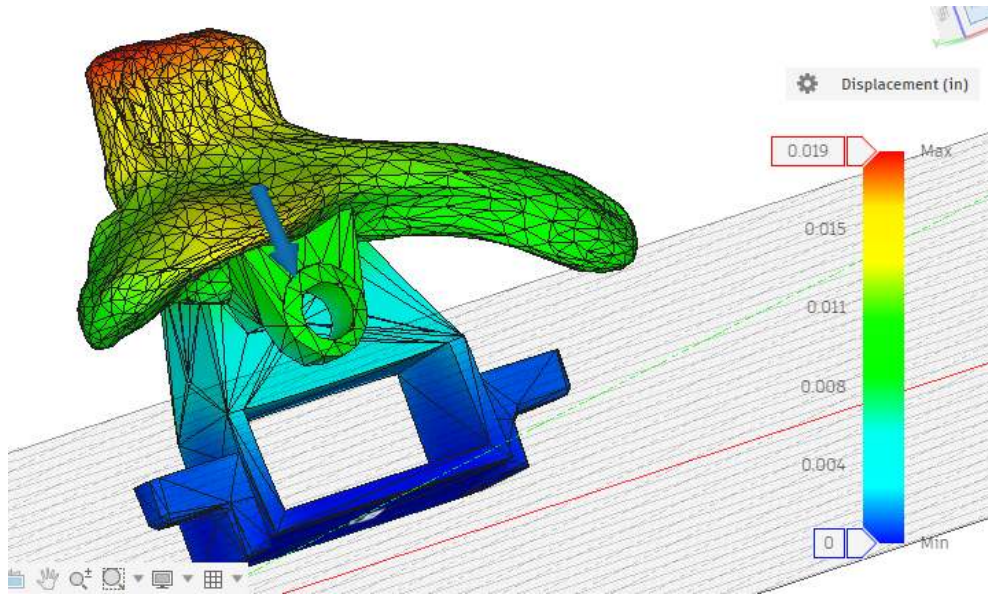


Figure 16: Hip Displacement

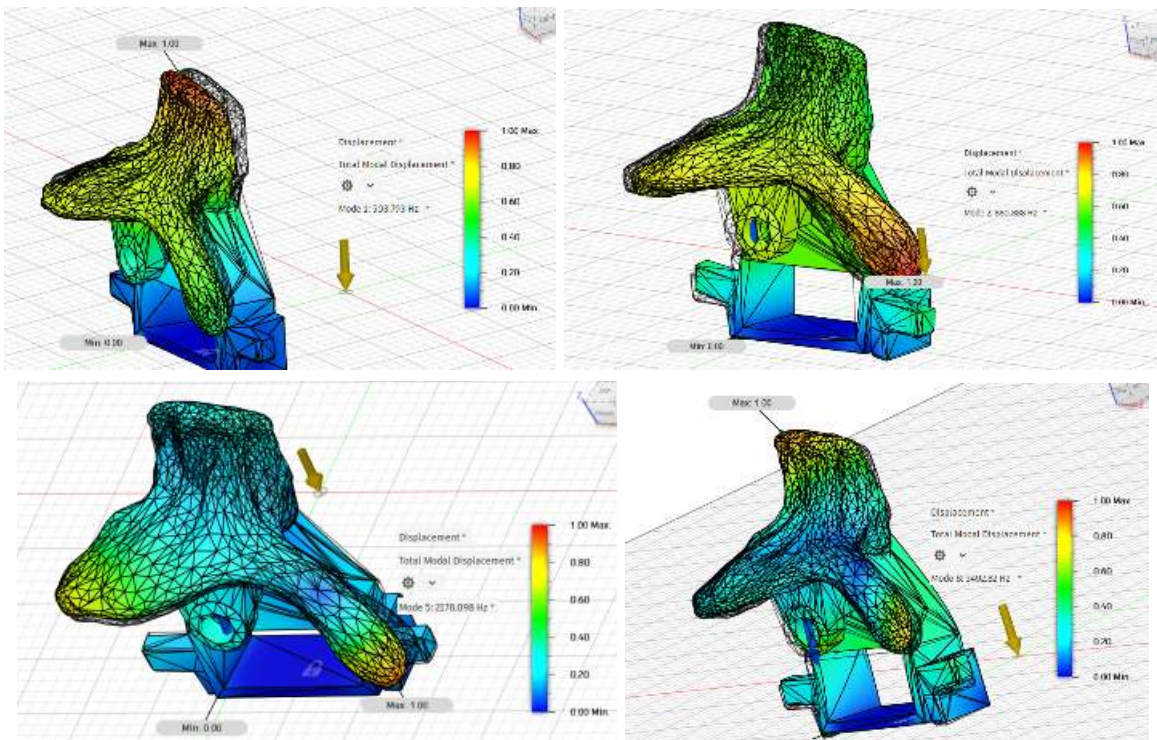


Figure 17: Mode Shapes from Vibration Analysis

Programming Layout

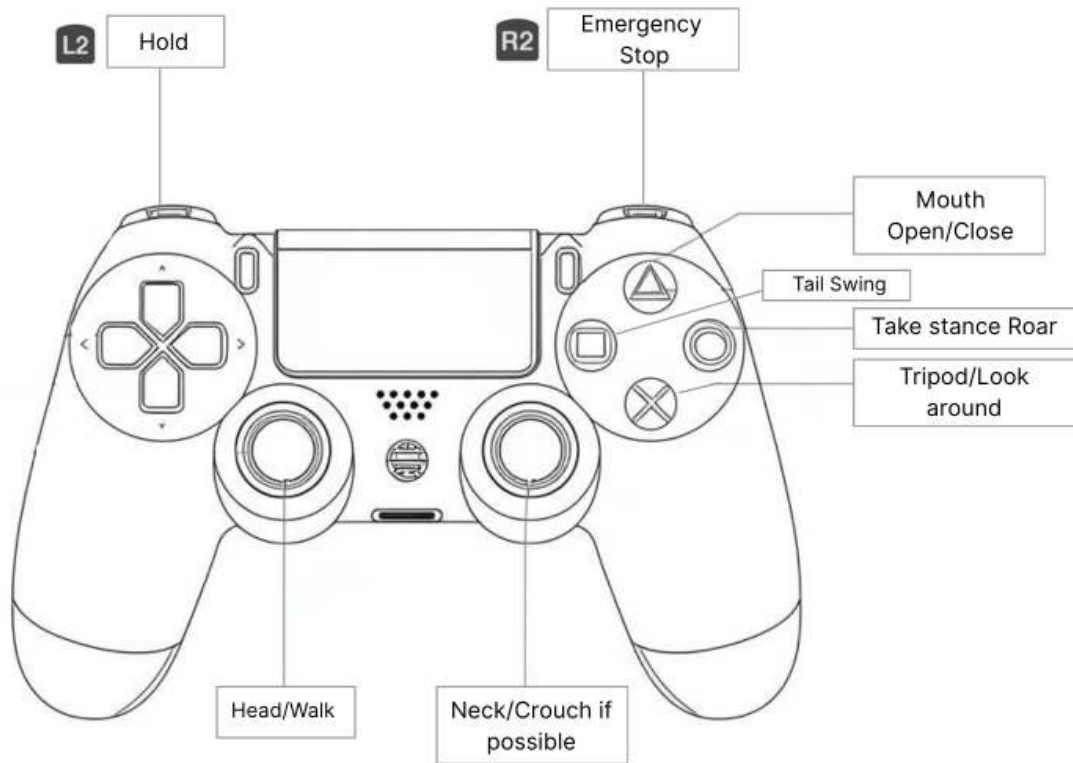


Figure 18: Proposed Controller Layout State 4

V. Prototyping and Testing

Prototyping is a necessary part of the engineering process. It allows the team to identify errors in the design before significant resources have been used. While comprehensive analysis and thorough project planning eliminate significant errors, even the most well-designed project will have discrepancies that are only apparent when a physical product is generated. Further, prototypes provide a physical proof of concept that can be presented to stakeholders or customers before the final product has been completed.

The purpose of the senior design course is for a team to generate a late-stage prototype, as opposed to a fully-developed product. However, for the team's internal goals, a sub-system prototype will also be developed. The team's final goal is to build a prototype of a stegosaurus animatronic with the capability to walk. Initially, the team will generate a prototype version of a single rear leg. This will allow for smaller-scale testing of the mechanical, electrical, and computer systems before the full-scale prototype is developed.

As the digital model is completed, analysis was run on the joints to determine their static and dynamic stresses. The team also generated a gait study, which allowed for the visualization of the walk-pattern of the stegosaurus. Using the gait study, the team was able to estimate the torque necessary for each joint. These calculations were validated once the team printed a model leg. Using a force-gauge, the force needed to actuate the joints was calculated and edits were made to the model.

The prototype will be tested against its design requirements. A test will be conducted against each requirement. The team will consider the prototype a success if this achieves all the principle design requirements.

VI. Development Planning

The key functional requirements of this project include an animatronic that is visually anatomically accurate to a stegosaurus skeleton. The animatronic must be able to walk, and therefore balanced dynamically. Additionally, the motion of the animatronic must reflect a plausible natural motion of the living stegosaurus. In terms of material functional requirements, the stegosaurus must be durable, and economical. Further, the material should be easily repairable. These requirements were decided by discussion with the team, advisor, museum staff, and a stegosaurus biomechanics expert.

Looking at the power requirements, the animatronic must have a reasonable runtime, and a reasonable recharge time. The control system must be able to coordinate between the mechanical and electrical components. The animatronic must be able to respond to user input.

The animatronic consists of three major systems and over 30 individual components. The goal of the team is to purchase all electrical components, while fabricating all mechanical components. Items, such as bearings, screws, or pins, will be purchased as well. This reduces the lead times of many items. Components with longer lead times include the physical structure, which must be 3D printed. Additionally, the code will take a significant amount of time to account for trouble-shooting and debugging. The electrical component with the longest wait time is the battery disconnect, which will take (3) weeks to be delivered. This part was ordered on

2/10/26. The team has no concerns about lead times causing conflicts with our development schedule.

The schedule has been instrumental in guiding the development of the project. From this point, all analysis has been completed, with the exception of a full-system stability analysis. This cannot be done until the full mechanical system has been assembled. The rear leg has been printed and wired, with the code operating the motors. In the coming weeks, the team will print the remaining three legs, assemble them, then wire the legs for power. Soon after, the pelvis, shoulders, spine, and head will be printed. The remainder of the body will be wired, and a full-system test can be run to integrate the code with the electrical. Once the code is operable, the gait study can be finalized, and ancillary movements for the head and tail can be generated and programmed. Finally, the team will ensure the prototype meets the design specifications and develop the user and maintenance manuals. A more complete path for development can be referenced in the master schedule provided below.

Appendices

I. Appendix A: Failure Modes and Effects Analysis

Structure								
Function	Failure Mode	Effect	Severity	Causes	Probability	Controls	Detection	RPN
Balance statically	Loses balance/doesn't balance	Robot falls over	3	weight imbalance, external strike, uneven ground	3	Internal gyroscope	1	9
Balance dynamically	Fails to balance during motion	Robot falls during motion	3	incorrect data input, gyroscope malfunction, joint malfunction	7	Microcomputer system check	3	63
Materials								
Function	Failure Mode	Effect	Severity	Causes	Probability	Controls	Detection	RPN
Joints and structures support weight	Robot too heavy	joint breaks, weight imbalance, energy inefficient, slow movement speed	5	internal parts too heavy, load placed on top of robot	3	scale	1	15
Parts are durable	Part breaks from material fatigue	Robot cannot walk, head or tail no longer move properly	7	overuse, external strain on parts	3	Factor of Safety	5	105
Use of 3D printing	Part breaks along layer line fault	legs/body stop functioning, Robot can no longer support weight	7	Shear force along printing layer line	3	infill density, Factor of Safety	2	42
Economical	Parts/manufacturing over budget	decreases potential customer base	1	design flaw	1	project budget, design choices	3	3
Economical	Product outside of consumer target range	decreases potential customer base	2	design flaw	1	project budget, design choices	3	6
Reliable	Part or parts of robot consistently break and/or require replacement	Maintenance cost increase, consumer interest decrease	5	Part weakness, shear forces, environmental damage	5	infill density, Factor of Safety, structural supports	3	75
Repairable	Robot maintenance excessively time consuming	Maintenance cost increase, consumer interest decrease	3	design flaws	2	design choices	3	18
Repairable/reliable	Robot maintenance excessively expensive	Maintenance cost increase, consumer interest decrease	5	design flaws, part choices	2	material/design choices, testing	2	20
Parts are durable	Parts experience severe deflection without breaking	Robot cannot walk, head or tail no longer move properly	7	overuse, external strain on parts	5	Factor of Safety	5	175

Replaceable	Robot maintenance excessively complex or consistently requires expert assistance	Maintenance cost increase, consumer interest decrease	7	part choice, design flaws	2	documentation, training, design choices	2	28
Power								
Function	Failure Mode	Effect	Severity	Causes	Probability	Controls	Detection	RPN
Runtime	Robot has low runtime	Use in exhibits is limited	5	battery capacity, excess weight, motor inefficiencies	6	battery choice, testing	3	90
Standards	Robot fails to meet a relevant FCC/Engineering standard	Lawsuit :(	9	design flaw, law oversight	1	research	5	45
Power distribution infrastructure	Wires break	Part or whole robot stops working until problem is fixed	9	Material fatigue, part interaction	3	design choices, testing	4	108
Power distribution infrastructure	Wires short	Potential fire hazard, risk of electrocution	9	wire breakage, water damage	3	design choices, testing	3	81
Overall function	Critical infrastructure not integrated properly	robot fails to work correctly	7	human error, broken/damaged parts	3	testing, visual verification	5	105
Weight	Power system overburdens robot/causes excessive strain on joints	Increased power usage, possible joint breakage	5	decreased runtime, battery strain	2	design choices, testing	2	20
Visible and audible communication of diagnostic data	Sound levels too loud	annoying to consumer	2	voltage to speaker too high, code problem	3	testing	1	6
Visible and audible communication of diagnostic data	Sound levels too soft	unusable to consumer, audio cue misinterpreted or missed	6	voltage to speaker too low, code problem	3	testing	1	18
Interference with mechanical articulation	something interferes with joint movement	Part or whole robot stops working, potential wire short	7	foreign objects, wires, internal defects	4	shielding, design choices	4	112
Rechargeable	Overcharging	potential battery explosion	9	code bug, microcontroller defect, voltage regulator defect	1	testing, voltage regulator	5	45
Rechargeable	Undercharging	potential battery explosion, decreased runtime	8	code bug, microcontroller defect, voltage regulator defect	1	testing, voltage regulator	5	40
Safety hazard	Battery puncture	Fire :(	10	external damage to battery	1	shielding	3	30
Software monitored	Software crashes	Robot stops working until problem is rectified	7	code bug, microcontroller defect	2	testing	6	84
Software monitored	Software receives bad sensor information	robot behaves erratically or not at all	6	external interference with sensor, damage to sensor, microcontroller problem, code bug	3	testing, calibration	5	90
Internal power source	Exceeds weight requirement	Increased power usage	4	human error	2	scale	2	16
Design error	Exceeds internal spatial dimensions	Less space for critical infrastructure, more vulnerable if not properly protected	4	human error	2	product description	3	24

Improper safety hazard	Malfunction damages other interior components	robot stops working or causes further damage to itself or environment	8	manufacturing defect, microcontroller problem, voltage regulator problem	2	testing of parts	1	16
Power system supports all requirements	Robot components exert electrical load above battery rating	battery strained beyond recommended use, damage or possible explosion	9	human error, design error	2	component amperage load testing	1	18
Articulation								
Function	Failure Mode	Effect	Severity	Causes	Probability	Controls	Detection	RPN
High resolution movement	Irregular or erratic movement	unpredictable or dangerous behavior	5	code error, joint problem, motor problem	5	testing, torque and ROM limitations	7	175
Degrees of freedom	Failure to move in a prescribed degree of freedom	partial or complete failure in locomotion	9	blocked joints, code error, mechanical failure, foreign object	4	testing, material and design choices	7	252
Realistic movement	Jerky or robotic movements	consumer immersion lost, interest decreases, potential loss of balance	4	blocked joints, code error, mechanical failure, foreign object	3	visual verification, gait study	7	84
Range of motion	Failure to move smoothly through entire range of motion	potential loss of balance/ability to move	1	blocked joints, code error, mechanical failure, foreign object	3	testing, gait study	3	9
Range of motion	Range of motion blocked by something	robot loses balance/ability to move	7	foreign debris	4	visual verification, shielding	3	84
Roar	Fails to produce sound	customer expectations not met	3	speaker defect, code error, microcontroller problem	2	test	1	6
Communication	Produces sound without movement	customer expectations not met, no movement	7	speaker defect, code error, microcontroller problem	2	test	1	14
Avoid obstacles	Hits obstacle	loss of balance, damage to robot or environment	8	operator failure, detection failure	3	test, training	1	24
Tail movement	Tail only partially moves	fails to meet standards	3	foreign object, material fatigue, broken part	3	test	2	18
Jaw movement	Jaw doesn't move or too much load	fails to meet standards	2	material experiences stresses past the yield point	3	tests, FEA	1	0
Tail movement	No tail movement at all	fails to meet standards	3	foreign object, material fatigue, broken part	2	test	2	12
Compact	Robot dimensions exceed requirements	problems with shipping and handling, excessive weight	5	design failure	2	size limitation	1	10
Control System								
Function	Failure Mode	Effect	Severity	Causes	Probability	Controls	Detection	RPN
Has a control system	loses control system	impossible to control, feedback loop or complete inaction	8	damage to control system, control system defect	4	shielding, integration	3	96
Responds to user input	User input misinterpreted by CS	erratic behavior	5	control system defect	2	Testing, component choice	3	30

Responds to user input	No response to user input/ blocked user input	erratic or no behavior	4	RC signal blockage, code issue, controller defect	4	testing, environment	3	48
Provides and stores diagnostic data	Reports inaccurate diagnostic data	erratic or no behavior	4	memory corruption, diagnostic tool damage/defect	3	testing, code debugging	2	24
Provides and stores diagnostic data	Fails to store diagnostic data	control system accuracy compromised	3	memory corruption, diagnostic tool damage/defect	3	testing, debugging	2	18
Pre-recorded movement	pre recorded movement does not play	breaks immersion, fails design standards	3	memory issue, code issue, microcomputer defect	2	testing, debugging, memory integrity validation	2	12
User-controlled individual motion	User error, robot piloted to dangerous area	robot breaks either structurally or electronically	8	operator	5	training, documentation	3	120
Makes noise/sound	audio files do not work	no sound, breaks immersion, fails design specifications	3	code problem, file corruption	2	testing	5	30
Makes no sound	audio devices broken or compromised	no sound, breaks immersion, fails design specifications	3	external damage, water damage, manufacturing defect	4	protection, testing	2	24
Easy to control	controls exceedingly complex	operator fails to properly operate robot	8	design flaw, human error	4	testing	3	96
Comes with instruction manual	instructions unclear	owner does not know how to use robot, potential unintended damage or failure to operate normally	8	design flaw, human error	4	Documentation, feedback	1	32
Lost instruction manual	Instruction manual lost, damaged, or destroyed	owner does not know how to use robot, potential unintended damage or failure to operate normally	8	shipping problem, user error	3	protection, documentation, redundancy (have more than 1 manual)	1	24
Balance feedback	Inaccurate data	Robot loses/fails to balance dynamically	6	sensor defect, code error, microcontroller problem	4	testing, sensor calibration	5	120
Wireless connection	Robot walks too far away	loses connection	3	environment	5	testing, automatic shutdown	4	60

Safety and Environment								
Function	Failure Mode	Effect	Severity	Causes	Probability	Controls	Detection	RPN
Standard atmospheric conditions	Excessive heat	electronics/material damage	8	environment	5	operating environment and safety constraints	1	40
Standard atmospheric conditions	Excessive cold	material damage	5	environment	5	operating environment and safety constraints	2	50
Standard atmospheric conditions	High humidity	electronics damage	8	environment	5	operating environment and safety constraints	5	200
Standard atmospheric conditions	Condensation	electronics damage	8	environment, rapid temperature change from hot to cold	5	operating environment and safety constraints	5	200
Automatic shut down	Signal failure	robot continues to operate outside of user definition	9	code error, microcontroller defect	2	test	2	36
Shut down on loss of communication	Signal failure	robot continues to operate outside of user definition	9	code error, microcontroller defect	1	test	1	9
Low emissions	Non-recyclable materials used	Environmental impact from end of product life cycle	4	material design choice	1	material design choices	1	4

Human interaction	Digits/body parts pinched in joints	medical injury, lawsuit	10	human error, insufficient safety protocol	6	operating environment and safety constraints	1	60
Human interaction	Jaw bite force causes soft tissue damage	medical injury, lawsuit	10	loss of control, human error (sticking hand into jaw), no detection system	2	safe distance from people, controlled by trained operator	1	20
Human interaction	Robot damaged or hit by human	robot functions compromised, possible need for repair.	7	operator error, environment, other human error	5	operating environment and safety constraints	1	35
Operator interaction	Does not respond to operator input	inactive or erratic behavior	7	control system error	3	testing feedback	1	21
Operator interaction	Operator improperly trained	Damage to robot or environment	6	human error, improper training	5	testing feedback	1	30
Operator interaction	Operator improperly trained	robot fails to move	5	human error, improper training	5	testing feedback	1	25
Operator interaction	Operator improperly trained	erratic behavior	5	human error, improper training	5	testing feedback	1	25

I. Appendix B: Functional Project Requirements

FUNCTIONAL REQUIREMENTS			
Req. #	REQUIREMENT	TARGET	VALIDATION METHOD
S	STRUCTURAL REQUIREMENTS		
F.S1	The animatronic shall balance statically.	The animatronic must be able to remain upright on its 4 limbs on a smooth surface within a range of +/- 10° in any cardinal direction.	Test: Stand animatronic on a surface, changing its degree of incline in the cardinal directions.
F.S2	The animatronic shall balance while walking.	Barring any external disturbances, animatronic must retain balance when cycling through all its gaits.	Test: The animatronic will cycle through all gaits on a smooth, flat surface.
F.S3	Joints and structure should support the weight and movement of the robot.	The animatronic shall not weigh more than 50 pounds, including all systems involved. The animatronic shall be able to support its entire weight.	Analysis: G force calculations. FEA.
M	MATERIAL REQUIREMENTS		
F.M1	Shall be durable.	Robot shall be able to withstand falls, small	Testing.

		drops, and regular handling.	
F.M2	Shall use additive manufacturing.	3D printing shall be considered for rapid prototyping and custom parts.	Material Analysis and Selection.
F.M3	Shall be economical	Material shall be economical to produce and use standard parts when possible.	Analysis on material costs.
F.M4	Shall be reliable.	Material can consistently withstand the loads and forces applied during operation.	Analysis.
F.M5	Shall be easily repairable.	Shall utilize sub-assemblies that are easily replaceable.	Analysis.
P	Power REQUIREMENTS		
F.P1	Power source shall have a long run time.	The robot runs for at least 30 mins before cutting off the battery.	Test.
F.P2	Power systems shall comply with all relevant FCC and Industry safety standards.	Refer to relevant standards.	Analysis, testing.
F.P4	Power distribution infrastructure will not interfere with mechanical articulation.	Joints will be able to have full range of motion without being impeded or damaging the Power system.	Test.
F.P5	The power system shall be light weight.	The Power system and battery combined must weigh less than 5 lbs.	Analysis.
F.P6	Battery swap out system	The battery shall easily be hot swapped for the next batter when the power shuts off..	Test.
F.P7	Battery shall be rechargeable.	There will be a USB C port on the back of the robot.	Visual.
F.P8	Software should monitor Power source health and fuel consumption.	The software should send the power data for data logging on the web server	Test.

F.P9	The Power system shall use an internal Power source.	the whole robot will function off the same single battery, subsystems will not have independent Power systems.	Visual.
F.P10	The Power system shall support all current requirements.	The main Power source shall be able to handle the max current loads from all internal electronics.	Test.
A	ARTICULATION REQUIREMENTS		
F.A1	Shall have a high resolution.	An individual joint should be able to stop and hold within 2 degrees of the desired angle.	Testing.
F.A2	Shall be flexible.	Each leg shall have a minimum of 2 degrees of freedom.	Analysis.
F.A3	Shall emulate realistic movement.	The general walking gait and structure will appear to move in a biological way.	Analysis on Gait.
F.A4	Shall have adequate range of motion.	Each joint and limb will move a max of 90 degrees and a minimum of 20 degrees to allow for full range of motion.	Analysis.
F.A5	Shall be able to ROAR.	The stegosaurus mouth will open at a 20° angle with respect to the top jaw with a tolerance of +/- 2°.	Test.
F.A6	Shall be able to turn to avoid obstacles.	The animatronic shall be able to turn at a 1:1 body length to turn radius ratio.	Test.
F.A7	Shall be able to move its tail.	Can wag tail at minimum of 45 degrees from the center axis.	Test.
F.A8	Shall be compact.	Wiring and circuitry shall be hidden in the main body of the robot.	Visual.
C	CONTROL SYSTEM REQUIREMENTS		
F.C1	Shall have a control system.	Shall be controlled by a single person.	Test.

F.C2	Shall respond to user input.	Control system shall have a low latency and update rate, with a command latency $\leq$ 100ms (Input to Response).	Test.
F.C3	Shall provide and store continuous diagnostics.	Stores Joint/Servo positions, IMU, Errors, Response Time, Controller Inputs, Power, and Temperature.	Demonstration.
F.C4	Shall have pre-recorded movement.	The interaction shall fully execute within 10 to 20 seconds then return to its idle state. The animation can be canceled mid execution.	Test.
F.C5	User shall be able to control individual actuator motion.	The controller will be able to independently control head motors, mouth motion, neck motors, and the tail motions along respective degrees of freedom. It will be able to hold the position within ± 2 degrees of the commanded orientation.	Test.
F.C6	Shall make noise/sound.	ROAR volume shall be at minimum 50 decibels.	Test.
F.C7	Movement and walking shall be easily controlled.	The controller shall control direction of gait and speed.	Test.
F.C8	Shall have an operators instruction manual.	Easy to operate and learn control mapping, with an included instruction manual.	Visual.
F.C9	Shall have a balance feedback system.	Stop animation and return to idle if animation is interrupted or loses balance.	Test.
F.C10	Shall be connected wirelessly.	Shall have at least a 50ft range.	Test.
E	SAFETY AND ENVIRONMENT REQUIREMENTS		
F.E1	Shall operate in dry conditions.	Robot shall not be submerged in water or exposed to rain.	Test.

F.E2	Shall operate in standard atmospheric conditions.	Standard atmospheric conditions are defined as a temperature of 15°C (59°F) and a pressure of 101,325 pascals (14.7 psi).	Test.
F.E3	Shall have an integrated shut down system.	Program will be able to sense if the robot falls over and turns off.	Test.
F.E4	Shall have a mechanical shut down system.	Will be able to turn off in less than 2 sec if a switch is pressed.	Test.
F.E5	Shall shut down if remote contact is lost.	Shall shut down within 5 seconds of loss of contact.	Demonstration.
F.E6	Shall have low emissions.	Shall not damage its surrounding environment or objects.	Analysis.

II. Appendix C: Component Project Requirements

COMPONENT REQUIREMENTS							
Req. #	System	Component	Stretch Goal	Requirement	Validation Method	Units	Range
C.C1	Control System	On/Off switch		Shall be easily accessible	Inspection	N/A	N/A
C.C2	Control System	On/Off switch		Shall immediately cut all Power to the robot.	Inspection	N/A	N/A
C.C3	Control System	Remote Transmitter/ Controller		Shall have a minimum range of [50] ft.	Testing	Ft	>= 50
C.C4	Control System	Remote Transmitter/		Shall stop all movement if the robot loses range or remote	Testing	N/A	N/A

		Controller		contact.			
C.C5	Control System	Controller		Will contain a computer that is able to be programmed allowing for precise control of joints and motion.	Testing	N/A	N/A
C.C6	Control System	Motor Controller		Will interface between the primary computer and the motors, combining positioning requests and handling motor electrical requirements.	Testing	N/A	N/A
C.C7	Control System	IMU	X	Will be able to keep track of inertia when the robot moves.	Testing	N/A	N/A
C.C8	Control System	Camera	X	Will be able to take in visual data which will affect how the robot moves.	Testing	N/A	N/A
C.C9	Control System	Gyroscope	X	Determine whether the position of the robot is upright or if it has fallen over.	Testing	N/A	N/A
C.C10	Control System	GPS	X	Determine where the robot is located in space.	Testing	N/A	N/A
C.C11	Control System	I/O		Input/Output connections to support everything and there are enough connections.	Analysis	N/A	N/A
C.E	Electrical System						
C.E1	Electrical System	Battery Pack		Shall have a run time of at least [30] min.	Test	min	>=30
C.E2	Electrical	Battery Pack		Shall charge in under [180] min.	Analysis	min	<=180

	System						
C.E3	Electrical System	Drive Motors		Shall have adequate torque in order to move each limb.	Analysis	N/M	N/A
C.E4	Electrical System	Drive Motors		Shall have adequate speed in order to replicate a realistic gate.	Analysis	Rad/Sec	N/A
C.E5	Electrical System	Animation Motors		Shall have between 2 to 18 lb.-rad ² torque in order to move non-structural elements.	Analysis	N/M	N/A
C.E6	Electrical System	Speaker		Will be able to communicate audio at a volume of at least [50] decibels.	Testing	dB	>=50
C.S	MAIN STRUCTURE						
C.S1	Main Structure	Power Module		Shall not exceed the physical dimensions of the robot.	Analysis	N/A	N/A
C.S2	Main Structure	Power Module		Shall be able to store and protect all main electrical components with the exception of motors.	inspection	N/A	N/A
C.S3	Main Structure	Power Module		Shall be impact resistant	Inspection	N/A	N/A
C.S4	Main Structure	Hardware/ Fasteners		Shall use off the shelf, metric Nuts & Bolts unless a specific part requires otherwise.	Inspection	N/A	N/A
C.S5	Main Structure	Hardware/ Fasteners		Shall use off the shelf standard size Bearings, Pins, & other hardware.	Inspection	N/A	N/A

C.S6	Main Structure	Bones		Main bone elements will and geometry will be within 20 percent of measurements taken from an existing scan.	Analysis	Precent	<= 20
C.S7	Main Structure	Back Bone		The main structure, spine & rib cage, shall be able to support 5 lb. weight.	Analysis	LBS	TBD
C.S11	Main Structure	Fins		Shall have fins along its spine, as the fossil record shows.	Analysis	Count	11
C.S12	Main Structure	Tail Spikes		Shall have tail spikes in accordance with the fossil.	Analysis	Count	4
C.S13	Main Structure	Feet		Interface material shall provide adequate friction to be able to have proper traction.	Analysis	N/A	N/A
C.J	JOINTS						
C.J1	Joints	Leg Joints		Each leg will have at least 2 degrees of freedom.	Analysis	N/A	N/A
C.J2	Joints	Leg Joints		Each joint shall be able to carry the weight of the robot.	Analysis	FOS	>=2
C.J3	Joints	Leg Joints		Each joint shall be able move 20 degrees in order to emulate movement.	Analysis	Degrees	TBD
C.J4	Joints	Neck		The neck will have a minimum segment of [1] moving parts or vertebrae.	Analysis	Count	>=3
C.J5	Joints	Tail		The tail will have a minimum segment of [1] moving parts or vertebrae.	Analysis	Count	>=8

C.J6	Joints	Mouth		The mouth shall be able to open at least [20] degrees.	Analysis	Degrees	≥ 20
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	Schedule
Week Of	Goals for the Week
Week 1: 1/12/26	- M4 Division of Work - Submit Budget to Stresau - Request Material From Inventory - Gait Study Started
Week 2: 1/19/26	- Draft gait study angle lists and animation - Work on model - Schedule finalization - Head and Tail models complete
Week 3: 1/26/26	- Finalize joint angle lists - Contact Matthew for input on joint angle animations - Begin torque calculations using gait study angles - Begin stress calculations using joint angles - Begin code development - Order Parts (Computers & Motors) - Model Material (Motors, Components) - Material acquisition

Week 4: 2/2/26	- Begin power calculations based on torque calculations - Complete torque calculations - Complete stress calculation - M5 Components distributed among team - Single rear leg model complete - Museum Project Update - Gear Ratios - Single leg components printed
Week 5: 2/9/26	- Work on M5 - Power system for single leg components wired - Code de-bugging for initial leg - Update Specifications
Week 6: 2/16/26	- Work on M5 - Single leg assemble to test: - Code - Joint Angles - Torque - Strength/Stability - 4 leg models complete
Week 7: 2/23/26	- Remaining 3 legs printed - Begin wiring power systems for remaining 3 legs - Code developed for 4 legs to be integrated together - Develop angle lists for head and tail movements - Add angle list to existing animation - Again, contact Matthew about project update - Body model complete
Week 8: 3/2/26	- Body printed - Remaining 3 legs assembled (mechanical and electrical) - Begin testing code integration of 4 legs - Gait studies for ancillary movements
Week 9: 3/9/26	- Head and Tail printed - Legs mounted to body - Balance testing - Debug code for 4 leg integration
Week 10: 3/16/26	- Wire power system for head and tail - Mount head and tail to body - Integrate code with head and tail movement
Week 11: 3/23/26	- Code ancillary movements - Test specifications from M2 - Write user manual
Week 12: 3/30/26	- Assign M6 presentation material
Week 13: 4/6/26	- Finish M6 material

Week 14: 4/13/26	- Finish M6 material
Week 15: 4/20/26	- Prepare for Showcase

Resources

- [1] P. Berkelman, P. Rossi, T. Lu, and J. Ma, "Passive Orthosis Linkage for Locomotor Rehabilitation," *2007 IEEE 10th International Conference on Rehabilitation Robotics*, Jun. 2007, doi:.
- [2] S. S. Bhavikatti, *Mechanics of Solids*. Newage International Publishers, 2000.
- [3] R. Richards, *Principles of Solid Mechanics*. CRC Press, 2000.